Referee Report: "Multiproduct Intermediaries"

Rhodes, Watanabe, and Zhou (2021)

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Introduction

Multiproduct intermediaries are still dominant even though manufacturers can now sell directly. The European Commission (2017) reports that in several EU sectors more than half of manufacturers sell direct online (85% in clothing), and OECD (2019) documents the same trend. At the same time Amazon's net sales exceed $500bn a year, large retailers keep broad assortments, and use of private labels and exclusive products is growing. So there is a tension that we need to understand: if direct channels are cheap, why do intermediaries thrive, and why do they rely more and more on exclusivity?

The paper gives a clean answer to this. In the standard story, intermediaries earn rents by reducing search costs or by lowering prices. Rhodes, Watanabe and Zhou show instead that an intermediary can be profitable purely through assortment choice — even when it is no more efficient at search, and even when prices stay at the monopoly level. The mechanism is a reallocation of search across products: by stocking some high-surplus products exclusively —for which the intermediary incurs in losses—, the intermediary draws consumers in, and then makes its money on the high-margin, low-surplus products it also carries, which consumers would not have searched for on their own.

The referee report is structured as follows: in section 1, I outline the bar-bones model they develop to understand this tension. In section 2, I outline some points of the model that deserve discussion. Finally, in the last section, 3, I propose a natural extension to one of the main criticism I outline above.

1 Model

There is a continuum of manufacturers of measure one, each producing a different product i with constant marginal cost $c_i \geq 0$ and demand $Q_i(p_i)$. A unit mass of consumers have identical preferences and additive utility across products, but differ in their unit search cost $s \sim F$ on $(0, \bar{s}]$. Visiting a manufacturer costs s; visiting the intermediary, which stocks a measure m of products,

costs $h(m)\,s$. There is a single intermediary with full bargaining power, who makes take-it-or-leave-it two-part-tariff offers (τ_i, T_i) and may demand exclusivity[1].

Since consumers cannot observe prices before searching, "hold-up" à la Diamond (1971) implies that every active seller of i charges the monopoly price p_i^m. Each product is then summarized by two numbers,

$$\pi_i \equiv (p_i^m - c_i)\, Q_i(p_i^m), \qquad v_i \equiv \int_{p_i^m}^{\infty} Q_i(p)\, dp,$$

i.e. the per-consumer monopoly profit and consumer surplus. Lemma 1 then says that the intermediary optimally sets $\tau_i = c_i$ (no double marginalization) and $T_i = \pi_i F(v_i)$, which is exactly the manufacturer's outside option when selling direct – consumers self-select into searching it iff $s \leq v_i$. The product space therefore collapses to $\Omega \subset \mathbb{R}_+^2$ with joint distribution $G(\pi, v)$, and this is the "workhorse" or pivotal part of the paper. This allows them to simplify a complex problem into a nicer and tractable setting.

With this result at hand, they will explore two cases:

"Bare-bones" case. Assume only exclusive contracts, $h(m) = m$, and no capacity constraint ($\bar{m} = 1$). Let $A \subseteq \Omega$ be the stocked set with measure $m = \int_A dG$. A consumer of type s visits the intermediary iff her expected surplus exceeds the search cost,

$$\int_A v\, dG \;\geq\; s \int_A dG \quad \Longleftrightarrow \quad s \;\leq\; \hat{v} \;\equiv\; \frac{\int_A v\, dG}{\int_A dG},$$

so a mass $F(\hat{v})$ of consumers visit. Stocking product (π, v) generates net profit

$$\pi\big[F(\hat{v}) - F(v)\big] :$$

the intermediary earns $\pi F(\hat{v})$ from the consumers it attracts, but must pay the manufacturer its outside option $\pi F(v)$. Products with $v < \hat{v}$ make money; products with $v > \hat{v}$ lose money. With a stocking indicator $q(\pi, v) \in \{0, 1\}$, the problem is

$$\max_q \int_\Omega q(\pi, v)\, \pi\big[F(\hat{v}) - F(v)\big]\, dG \quad \text{s.t.} \quad \hat{v} = \frac{\int_\Omega q\, v\, dG}{\int_\Omega q\, dG}.$$

Treating the constraint with multiplier λ, the Lagrangian gives the value of stocking (π, v) as

$$\underbrace{\pi[F(\hat{v}) - F(v)]}_{\text{direct effect}} + \underbrace{\lambda(v - \hat{v})}_{\text{indirect effect}},$$

which is the central trade-off of the paper. The direct effect is the within-product profit; the indirect effect captures how the product shifts $\hat{v}$ and so the visiting mass. The two have opposite signs (since $\lambda > 0$ at optimum), and the optimal assortment ends up with two negatively correlated

[1] Note that timing is the following: (1) contracting, (2) pricing, (3) consumers observe *availability* (not prices), search sequentially, and buy.

regions: among high-v products, the intermediary picks the low-π ones (cheap traffic generators), and among low-v products it picks the high-π ones (profit generators). This is what keeps the intermediary profitable even when $h(m) = m$.

Note that the asymmetry that makes all of this work is one-stop shopping: searching n manufacturers costs ns, but the intermediary is reached with a single $h(m)s$ regardless of how many products it carries. A consumer who would not pay s to look up some low-v product on its own will buy it once she is already inside. Take this asymmetry away and the indirect effect vanishes; the intermediary cannot exist profitably.

General case. With non-exclusive contracts, a general $h(m)$, and a capacity constraint $m \leq \bar{m}$, the structure and resylts of the bare-bones model carry over: a search threshold $\hat{s}$ replaces $\hat{v}$ and Proposition 2 preserves the same negatively correlated pattern. With economies of search ($h'(m) < 1$), the intermediary keeps all high-v products; with diseconomies ($h'(m) > 1$), only low-π ones survive. Larger capacity means a larger but less exclusive store.

Applications. The paper applies the framework to two settings. For DTC sales, the model rationalizes a clean comparative static: as manufacturers' outside options rise, the intermediary should shrink its assortment and rely more heavily on exclusivity. For shopping malls, the framework explains why anchor stores are subsidized to enter (they are loss-making traffic generators) and recovered against by charging higher rents to non-anchor stores, who benefit from the traffic.

2 Comment-discussion

The model is elegant and captures a key tension of the empirical setting. However, a few assumptions deserve comment:

Single intermediary. A single intermediary with full bargaining power captures the entire traffic-generation rent. This is restrictive in a way that matters for the paper's main predictions: in most realistic settings (online retail, OTAs, streaming, malls in dense areas) intermediaries actually compete, and that competition should change who captures the rent. With competing platforms, a high-v manufacturer's outside option is no longer just direct sales — it can threaten to go to the rival — and rents should shift in its direction. That in turn weakens the DTC story: if manufacturers already extract rent from inter-platform competition, easier direct sales is a smaller marginal change than the model suggests. The paper flags this as future work and notes that the (π, v) approach may not extend cleanly. I sketch below what I think survives.

One-stop shopping. The asymmetry between $h(m)s$ at the intermediary and ns across manufacturers is not just one assumption among several — it is the key mechanism. Without it, the indirect effect $\lambda(v - \hat{v})$ collapses, and the intermediary needs either better search efficiency or lower prices to exist. So the question of when this reduced form is appropriate is essentially the question of when the model applies at all. For a mall or department store, where a single trip really does cover

everything inside, it is natural. For online platforms it is less obvious: in-platform search often looks closer to per-product, so $h(m)$ should grow at least linearly. Corollary 1 partially addresses this (only low-π traffic generators survive when $h'(m) > 1$), but a single fixed cost of "entering" the platform remains, and that fixed cost is doing a lot of the work. The paper could be more explicit about when the reduced form fits.

Demand heterogeneity. Consumers differ only in s, and this is precisely what delivers uniform monopoly pricing across channels and makes the (π, v) representation work in the first place. Section 7.2 introduces some demand heterogeneity, but only of a form (quantity scaling) that preserves uniform pricing by construction — footnote 47 acknowledges this is the binding restriction. Once consumers have channel-specific tastes or differ in which products they actually want, the manufacturer's optimal direct price and the intermediary's optimal price diverge, and a product can no longer be summarized by two numbers. This matters substantively: in practice consumers do not all want every product, and platforms attract different consumer mixes. The polarized (π, v) assortment pattern — low-π high-v traffic generators paired with high-π low-v profit generators — is a prediction conditional on the framework's homogeneity assumption, and it is not obvious how robust it is to even mild taste heterogeneity.

3 Extension sketch: competing platforms

Take the market-structure point in its narrowest form. Two symmetric platforms $j \in \{1, 2\}$ bid simultaneously for exclusivity on a single product (π, v), holding the rest of each assortment fixed. Stocking it changes platform j's profit by

$$\pi[F(\hat{v}_j) - F(v)] + \lambda_j(v - \hat{v}_j),$$

the same direct + indirect decomposition as in the baseline. With Bertrand-style bidding for exclusivity, the winning platform's transfer is bid up to the loser's reservation value, so the manufacturer's transfer rises from $\pi F(v)$ to roughly $\pi F(v) + \lambda_{-j}(v - \hat{v}_{-j})^+$.

Two things follow, conjecturally. The rent on high-v products shifts from the platform to the manufacturer, since these are exactly where the indirect value is large and Bertrand competition bites. The low-v, high-π profit-generator region is roughly untouched, since nobody bids for those as traffic generators. The polarization story should survive, with redistributed rents.

The asymmetric case is harder to pin down. One natural intuition is that a larger platform has a bigger λ and so bids more aggressively for high-v exclusives, predicting that larger platforms hold proportionally more exclusive high-v inventory. A different intuition, which I think is closer to what we see, is that platforms specialize horizontally (luxury vs. mass-market, generalist vs. niche) and so are not really bidding for the same products at all. This is exactly the kind of asymmetry the (π, v) framework cannot accommodate, for the reason discussed above: it requires consumer types that differ in tastes, not just in s. Either way, once visiting thresholds depend on rival assortments the consumer problem stops being separable, and existence is no longer obvious.